Think Data, Think Artha.

Enhancing Patient Record Deduplication in Diabetes Care with ML Classification Models

Case Study

Estimated Annual Business Impact

$700K+	25-40%	60%
Annual cost savings via reduced redundant labs, admin time, and claim rejections.	Improvement in care quality through unified diabetes patient views.	Reduction in admin workload through advanced automation

Business Problem

In healthcare, particularly in chronic disease management like diabetes, maintaining accurate and consolidated patient records is critical for diagnosis, treatment planning, risk stratification, and compliance. However, healthcare systems often struggle with:

Duplicate patient records across EHR systems, labs, and outpatient facilities due to inconsistent data entry or system silos.

Up to 10–20% of records being redundant or fragmented, leading to poor clinical decision-making.

Manual deduplication increasing administrative workload and risking HIPAA violations if mismatches occur.

Inconsistent records leading to missed diagnosis patterns, duplicate testing, and suboptimal care for diabetic patients.

High-Level Solution

ML-BASED RECORD MATCHING VIA CLASSIFICATION

- Formulated deduplication as a binary classification problem — predicting whether two records refer to the same patient.
- Utilized similarity metrics on names, addresses, date of birth, lab results, and ICD codes as input features.
- Trained classifiers (e.g., Logistic Regression, Random Forests) on labeled match/non-match pairs for high precision.

HUMAN-IN-THE-LOOP WITH CLINICAL VALIDATION

- Uncertain matches were reviewed by clinical data stewards to ensure patient safety and regulatory accuracy.
- Models retrained iteratively based on validated examples to improve performance over time.

UNIFIED DIABETES PATIENT RECORD CREATION

Merged identified duplicates into single longitudinal records enabling better glycemic trend analysis, prescription adherence tracking, and population health reporting.

TECHNOLOGY STACK

- Python + scikit-learn: Classification model training and scoring.
- HL7 FHIR-based APIs: Standardized data exchange and identity resolution.
- Databricks + Delta Lake: Unified data processing across hospital networks.
- Streamlit UI: Review console for clinical operations.
- EHR Integration (Epic/Cerner): Real-time updates to patient records.

Healthcare Impact Metrics

Impact Area	Before	After	Business Impact
Duplicate Patient Records (Diabetes)	10–20%	<2%	↑ Reliable, single view of diabetic patients
Missed Diagnosis & Lab Overlap	Frequent	Rare	↓ Redundant testing and missed trends
Manual Record Review Time	60–100 hrs/month	<25 hrs/month	↓ 60% reduction in admin workload
Clinical Decision Support Accuracy	Impaired by fragmented data	Enhanced	↑ Better insulin/treatment guidance
Regulatory Risk (HIPAA, Safety)	High	Reduced	↑ Compliance and patient trust
Patient Outcome Monitoring	Incomplete histories	Longitudinal tracking	↑ Improved outcomes & readmission reduction

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